Market Makers' Charm Exposure Projection

Why track charm exposure

Charm is one of the key "second-order Greeks" used to understand how the delta of an option changes over time. While many traders focus heavily on gamma exposure, they often overlook other influential Greeks like charm, which plays a critical role in market makers' (MMs) delta hedging activities.

Unlike parameters like price and implied volatility (IV), which can fluctuate both up and down, time only moves forward. This inevitability makes charm a persistent and somewhat predictable force, often compared to a steady wind that passively pushes changes in delta. It is this predictability that makes it crucial for understanding market maker hedging activity.

What Does Charm Measure?

Charm measures how the delta of an option changes with the passage of time, assuming all other variables (like price and volatility) remain constant. Using the Black-Scholes-Merton (BSM) model, some key dynamic influences on delta are:

- **Gamma**: Changes in delta due to movements in the underlying price.
- **Vanna**: Changes in delta due to shifts in implied volatility.
- **Charm**: Changes in delta due to the passage of time.

As time passes, the delta of OTM options naturally shift.

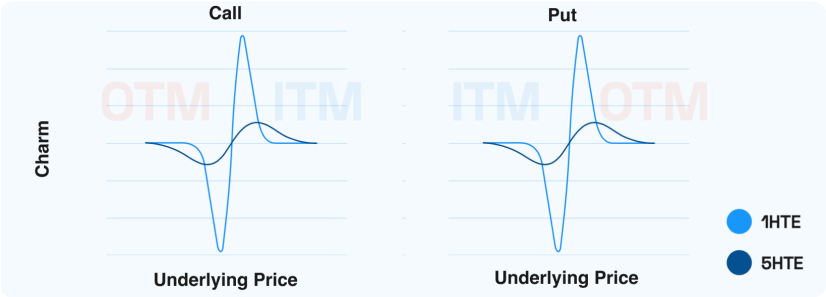

Figure 1: charm graph vs spot here

The impact of charm grows stronger as options approach expiration, a phenomenon amplified by the recent rise in 0DTE (zero days to expiration) options.

A modern approach: Our MM charm exposure heatmap

In the spring of 2022, we introduced a pioneering approach to address the limitations of traditional methods used to assess market makers' charm exposure. Our heatmap offers a more accurate and dynamic representation by calculating the entire charm exposure of market makers' portfolios and projecting it throughout the trading session. Unlike conventional linear models, our solution provides a surface map that enhances traders' contextual understanding of market dynamics.

A key innovation is our reliance on actual market maker option positions rather than assumption-based models. This ensures a precise assessment of their positioning, enabling traders to gain actionable insights. Intraday projections further enrich this analysis by providing time-sensitive context, particularly leveraging the mechanics of 0DTE options.

The heatmap computes the exposure of all options across all expirations, delivering a comprehensive and accurate outlook. By default, positive charm is represented in orange, while negative charm is shown in blue, making it easy to interpret the directional influence.

As the trading session progresses, heightened convexity becomes evident on the heatmap, primarily driven by the increasing influence of 0DTE options as they near expiration. This evolving dynamic offers traders a clear visual understanding of how charm exposure intensifies and shifts throughout the day.

The dynamic nature of hedging

Market makers' hedging is a constantly shifting process, with multiple players competing in real time. Our tool captures this broader context by visualizing the complete charm exposure landscape, the effect of all options, strikes and expirations combined

Visualization

We offer multiple ways to explore charm exposure:

- **3D Surface:** This visualization shows a surface map, with yellow markers representing the closing prices of candlesticks for your chosen timeframe
- **Heatmap:** A flattened version of the 3D surface for candlesticks overlay.

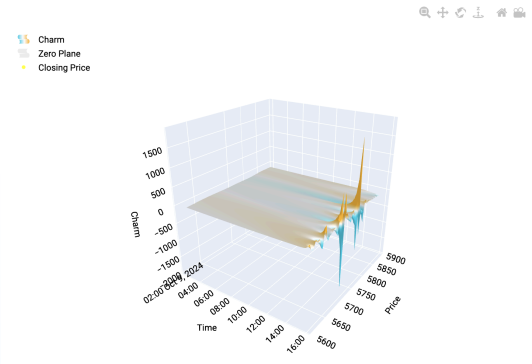

Figure 2: 3D Charm Surface

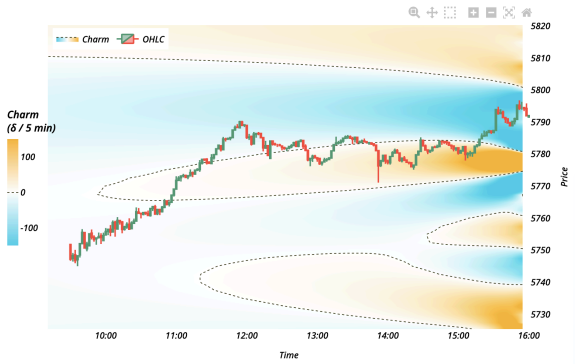

Figure 3: Charm heatmap

Interpreting Charm Exposure

Charm exposure significantly affects market makers' hedging behavior as delta exposure shifts over time. These changes create either suppressive or supportive effects on the price of the underlying asset, as market makers need to passively buy or sell the underlying asset to re-hedge their options portfolio.

Positive Charm – Suppressive

Positive charm occurs when the delta exposure of their options portfolio increases over time. To hedge this, market makers sell the underlying asset, such as ES futures for SPX options. This sustained selling pressure acts as a suppressive force on upward price movements, contributing to a gradual dampening of price momentum.

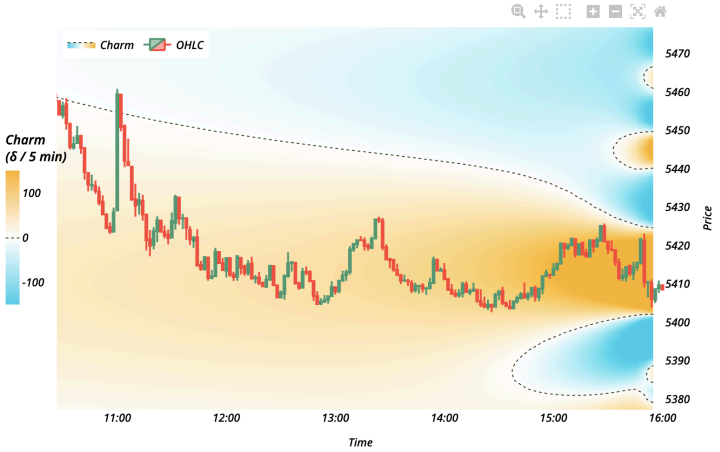

Figure 4: Suppressive charm

Negative Charm – Supportive

Negative charm happens when the delta exposure of their options portfolio decreases over time. In this scenario, market makers hedge by buying the underlying asset passively. This creates consistent buying pressure, which supports prices and mitigates downward movement.

Figure 5: Supportive charm

Charm as a Persistent Passive Force

Charm's magnitude is often mild but persistent, acting as a passive force influencing price dynamics over time. Think of it as a steady wind that, while subtle, can accumulate noticeable effects on the trajectory of the underlying price. For example, during periods of high suppressive charm, market makers' gradual selling can suppress an underlying rally, even when broader bullish sentiment prevails. Similarly, during periods of supportive charm, their passive buying can provide underlying price support amidst general market weakness. When charm aligns with broader market trends, it can act as a reinforcing force, amplifying either upward or downward movements, making trends more pronounced.

Charm Influence on Pinning

Pinning is the result of a multifactorial interplay, including customer flows and market makers' hedging dynamics, rather than a deliberate effort to anchor prices at the "max pain" level, as is often misconceived. Instead, it emerges naturally from the complex interactions of multiple factors.

Charm, ***while not the sole determinant***, can play a notable role in the pinning process when its magnitude is elevated. In an environment dominated by positive (suppressive) charm, prices may experience downward pressure, gradually moving toward what can be described as the "charm equilibrium point." Conversely, in a negative (supportive) charm environment, prices may drift upward, similarly converging toward this equilibrium.

The charm equilibrium point marks the transition, or "flip," between suppressive and supportive charm. At this point, market makers' hedging requirements achieve balance, reducing the need for passive buying or selling of futures to manage delta exposure over time. This equilibrium reflects a state where charm's directional influence is neutralized, providing a stabilizing effect on the price movement of the underlying asset.

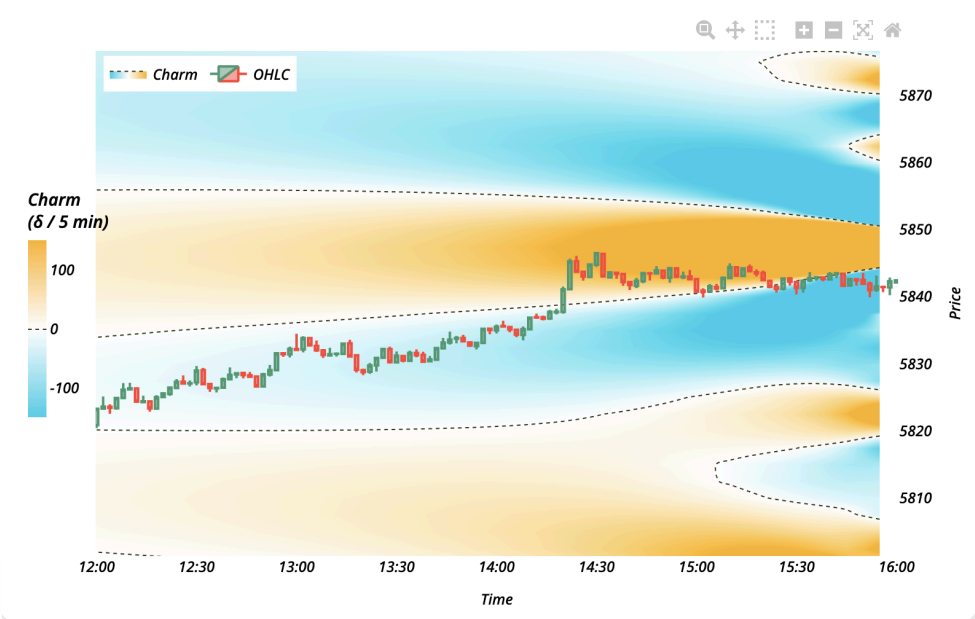

Figure 6: Charm contributing to pinning